

# Advanced Cryptography

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Differential Input Attacks — Countermeasures and NNT

# ROADMAP

## Countermeasures

**Signal to Noise Ratio**

Estimating the NNT

Impact of hiding countermeasures

Impact of secret sharing

# THREE PRONGED APPROACH

Protocol level: change key often

**Hiding:** remove link between values stored in the device and leakage; can be done by modifying hardware or software.

**Masking:** remove link between predictable intermediate values and values that are actually held in the device; can be done in hardware or software.

Hiding is often less effective than masking. To understand why, we need to understand how countermeasures impact on the number of traces, that are needed for a “successful” attack.

# CHARACTERISING SUCCESSFUL ATTACKS

Success is often understood as “percentage of attacks where the correct subkey is top ranked”.

The “rank” of a subkey is the position in an ordered list of subkeys, where the one with the highest e.g. correlation score is ranked top.

There are many factors that are taken into account in practice:

- ▶ How many traces are needed to ensure that  $x\%$  of attacks rank the correct subkey on top?
- ▶ How large is the key guess that is needed in an attack?
- ▶ How much effort (traces, time) went into finding a good model?
- ▶ How much effort went into finding an attack vector?

In the following, we will only focus on the number of needed traces (NNT).

# CHARACTERISING SIDE CHANNEL MEASUREMENTS

I now focus on power/EM. A similar argument can also be made for timing side channels.

Each measurement point is a function of the leakage of the associated operation (instructions), the data, and other, independent processes:

$$P_{total} = P_{op} + P_{data} + P_{noise}. \quad (1)$$

We typically think of the three components as “signal” (i.e. something that is data dependent) and “noise”. The “noise” is understood to be statistically independent of the “signal”.

An adversary “models” the signal: typically, they cannot capture all data-dependency.

## SIGNAL TO NOISE RATIO (SNR)

In the case of power/EM, the SNR is typically defined as ratio between “signal power” and “noise power”.

The power  $p(t)$  of an (electrical) signal (this includes the part of the signal that we call the noise) is defined as the product of current  $i(t)$  and voltage  $v(t)$ , i.e.  $p(t) = i(t) \cdot v(t)$ . Ohm's law relates voltage to current and resistance  $R$ , so  $v(t) = R \cdot i(t)$ . Therefore the power can be expressed as  $p(t) = i(t)v(t) = R \cdot i^2(t)$ .

Because of the data dependency, and the fact that we always have noisy observations, an SNR estimate relies on the **average** signal power:  $E[p(t)] = R \cdot E[i^2(t)]$ .

If we make measurements “mean free”, then  $E[i(t)^2] = \text{Var}(i(t))$ .

**The power of a signal can be estimated by estimating its variance.**

## SIGNAL TO NOISE RATIO (SNR), CONT.

### SNR

In (statistical) signal processing, the **signal-to-noise ratio (or SNR)** is defined as the ratio of the average power of a signal (meaningful information) to the average power of background noise at some point  $t$ :

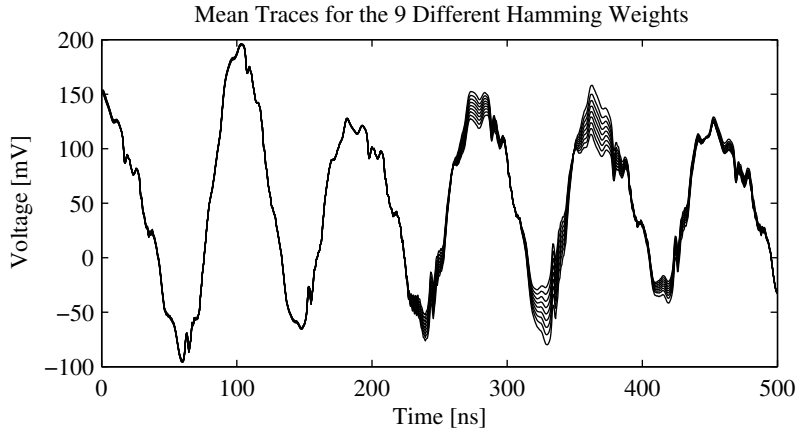
$$SNR(t) = \frac{Var(Signal(t))}{Var(Noise(t))}$$

In case of a differential style attack, the signal corresponds to what the adversary predicts, and therefore exploits. This may sometimes only be a fraction of  $P_{data}$ . We make this explicit and use the notation  $P_{exp}$ . The specific SNR then for a differential attacker is:

$$SNR(t) = \frac{Var(P_{exp}(t))}{Var(P_{noise}(t))} \quad (2)$$

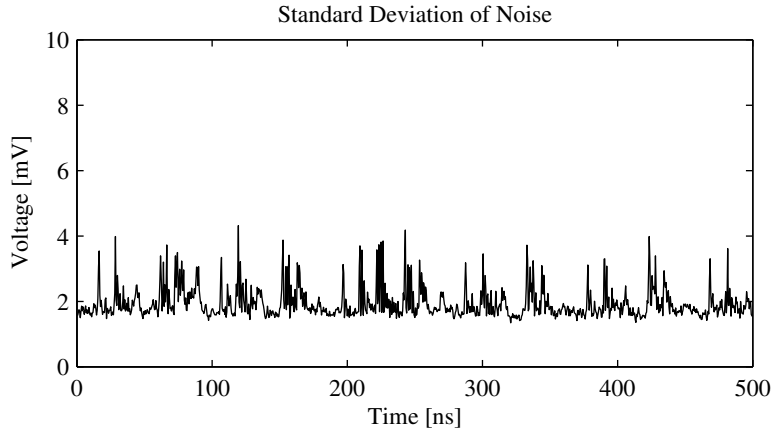
## SNR EXAMPLE

We estimate the average signal according to a given power model (shown in the figure below), and then estimate the variance across the groups.



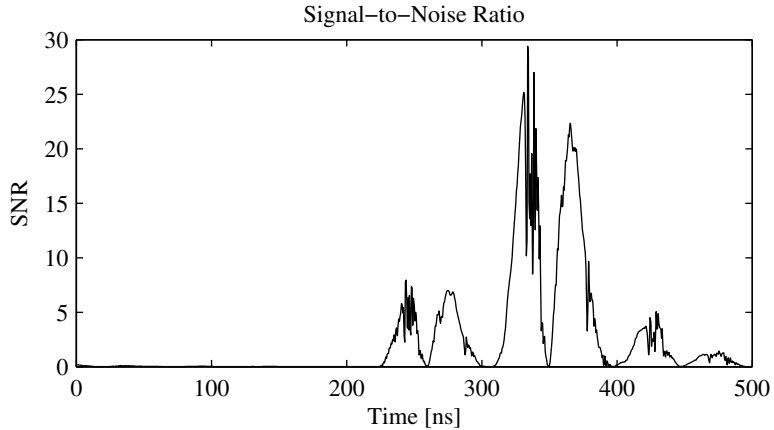
## SNR EXAMPLE, CONT.

We estimate the noise component via averaging over the “within group” variance.



## SNR EXAMPLE, CONT.

Now we look at the ratio.



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## SNR AND CORRELATION

The Pearson correlation coefficient  $\rho$  is essentially a “normalised” version of the covariance (i.e. it can only take values between plus and minus one:  $-1 \leq \rho \leq 1$ ). Because we do know the distributions that we are working with, we can only estimate the correlation from data, using the estimator  $r$  (see below).

In a differential input attack, we often use  $r$  to create scores for subkeys, by which we then rank the subkeys.

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X) \cdot Var(Y)}} \quad (3)$$

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (4)$$

## NUMBER OF NEEDED TRACES (NNT)

Any NNT estimate is specific to an attack configuration. A widely used attack configuration is based on using  $r$  with some model for the device under attack.

Because  $r$  is estimated from some data (the measurements), different sets of measurements will give different estimates for  $r$ . If we would repeat an attack many times, then the different outcomes for  $r$  would in fact look vaguely normally distributed.

So the distribution of  $r$  (at least if  $r$  is small) is well defined, and therefore we can work out its expectation and variance. With this we can draw on a whole lot of statistical machinery to work out an estimate for the NNT.

$$n = 3 + 8 \cdot \frac{z_{1-\alpha}^2}{\left( \ln \frac{1+\rho_0}{1-\rho_0} - \ln \frac{1+\rho_1}{1-\rho_1} \right)^2} \quad (5)$$

## RULE OF THUMB FOR NNT

If we may assume that the correlation for incorrect key candidates is close to zero, i.e.  $\rho_0 = 0$ , then we can simplify this further.

### Rule of thumb

The number of traces  $n$  that are needed to mount a successful DPA attack can be approximated as follows:

$$n = 3 + 8 \frac{z_{1-\alpha}^2}{\ln^2 \frac{1+\rho_{ck,ct}}{1-\rho_{ck,ct}}} \quad (6)$$

Because  $\ln(1 \pm x) \approx \pm x$ , we can turn this into even more of a rule of thumb:  $n \approx \frac{1}{\rho^2}$ .

## SNR AND CORRELATION

$$\begin{aligned}\rho(H_i, P_{total}) &= \\&= \rho(H_i, P_{exp} + P_{noise}) \\&= \frac{E[H_i \cdot (P_{exp} + P_{noise})] - E[H_i] \cdot E[P_{exp} + P_{noise}]}{\sqrt{\text{Var}(H_i) \cdot (\text{Var}(P_{exp}) + \text{Var}(P_{noise}))}} \\&= \frac{E[H_i \cdot P_{exp} + H_i \cdot P_{noise}] - E[H_i] \cdot (E[P_{exp}] + E[P_{noise}])}{\sqrt{\text{Var}(H_i) \cdot \text{Var}(P_{exp})} \sqrt{1 + \frac{\text{Var}(P_{noise})}{\text{Var}(P_{exp})}}} \\&= \frac{E[H_i \cdot P_{exp}] - E[H_i] \cdot E[P_{exp}]}{\sqrt{\text{Var}(H_i) \cdot \text{Var}(P_{exp})} \sqrt{1 + \frac{\text{Var}(P_{noise})}{\text{Var}(P_{exp})}}} \\&= \frac{\rho(H_i, P_{exp})}{\sqrt{1 + \frac{1}{\text{SNR}}}}\end{aligned}\tag{7}$$

If  $\rho(H_i, P_{exp}) = 1$ , then  $\rho^2 \approx \text{SNR}$ .

## SNR AND CORRELATION, CONT.

Now we can bring this observation together with our number of traces estimate from before, and we find that:

$$n \approx \frac{1}{\rho_{ck,ct}^2} \approx \frac{1}{\text{SNR}} = \frac{\text{Var}(P_{\text{noise}})}{\text{Var}(P_{\text{exp}})}. \quad (8)$$

- ▶ Halving  $\rho_{ck,ct}$  (aka SNR) means that four times more traces are needed.
- ▶ Reducing the signal/increasing the noise equally impacts on  $n$ .

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## HIDING VIA RANDOM DELAYS OR SHUFFLING

Hiding techniques that work based on inserting random delays (or random instructions/sequences of instructions) or re-ordering of instructions require a bit of thinking.

The correlation between the hypothetical power consumption for the correct key hypothesis  $H_{ck}$  and the power consumption at the moment of time  $ct$  is given by  $\rho_{ck,ct}$  which corresponds to  $\rho(H_{ck}, P_{total})$ .

To account that for some traces at time point  $ct$  something “random” is happening, we add a “hat”:  $\hat{P}_{total}$ . We use  $\hat{p}$  to quantify the probability that the “right” thing happens in time point  $ct$ , and therefore  $1 - \hat{p}$  is the probability that something “random” is happening.

$$Cov(H_{ck}, \hat{P}_{total}) = \hat{p} \cdot Cov(H_{ck}, P_{total}) + (1 - \hat{p}) \cdot Cov(H_{ck}, P_{other})$$

## WORKING OUT THE CORRELATION

We can plug this covariance into the definition for  $\rho(H_{ck}, P_{total})$  as shown in (9) and simplify the expression because  $Cov(H_{ck}, P_{other}) = 0$ , and  $P_{other}$  and  $P_{total}$  are independent.

$$\begin{aligned}\rho(H_{ck}, \hat{P}_{total}) &= \frac{\hat{p} \cdot Cov(H_{ck}, P_{total}) + (1 - \hat{p}) \cdot Cov(H_{ck}, P_{other})}{\sqrt{Var(H_{ck}) \cdot Var(\hat{P}_{total})}} \\&= \frac{\hat{p} \cdot Cov(H_{ck}, P_{total})}{\sqrt{Var(H_{ck}) \cdot Var(\hat{P}_{total})}} \cdot \sqrt{\frac{Var(P_{total})}{Var(P_{total})}} \\&= \rho(H_{ck}, P_{total}) \cdot \hat{p} \cdot \sqrt{\frac{Var(P_{total})}{Var(\hat{P}_{total})}}\end{aligned}\tag{9}$$

## WHAT IF AN ADVERSARY ACCOUNTS FOR MISALIGNMENT?

$$\begin{aligned}\rho(H_{ck}, \sum_{i=1}^l P_i) &= \frac{E(H_{ck} \cdot \sum_{i=1}^l P_i) - E(H_{ck}) \cdot E(\sum_{i=1}^l P_i)}{\sqrt{\text{Var}(H_{ck}) \cdot (\text{Var}(\sum_{i=1}^l P_i))}} \\&= \frac{E(H_{ck} \cdot P_1 + H_{ck} \cdot \sum_{i=2}^l P_i) - E(H_{ck}) \cdot (E(P_1) + E(\sum_{i=2}^l P_i))}{\sqrt{\text{Var}(H_{ck}) \cdot \text{Var}(P_1)} \sqrt{\frac{\sum_{i=1}^l \text{Var}(P_i)}{\text{Var}(P_1)}}} \\&= \frac{E(H_{ck} \cdot P_1) - E(H_{ck}) \cdot E(P_1)}{\sqrt{\text{Var}(H_{ck}) \cdot \text{Var}(P_1)} \sqrt{\frac{\sum_{i=1}^l \text{Var}(P_i)}{\text{Var}(P_1)}}} \\&= \frac{\rho(H_{ck}, P_1)}{\sqrt{\frac{\sum_{i=1}^l \text{Var}(P_i)}{\text{Var}(P_1)}}}\end{aligned}\tag{10}$$

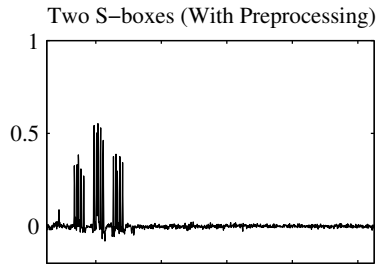
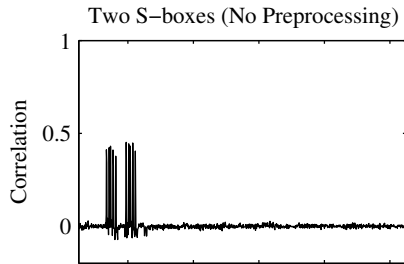
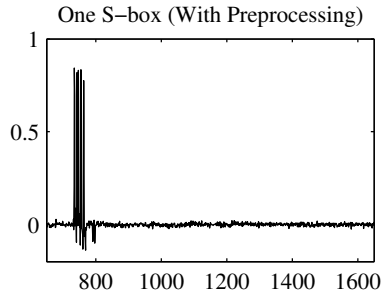
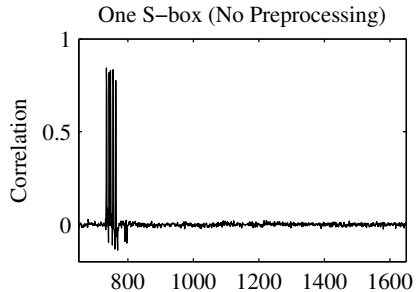
## I

If the variances in the trace points are roughly equal, then we can simplify this expression further and it becomes

$$\rho(H_{ck}, \sum_{i=1}^l P_i) = \frac{\rho(H_{ck}, P_1)}{\sqrt{l}}. \quad (11)$$

This last equation tells us that the impact of misalignment on the correlation is that it reduces it by a factor that corresponds to the square root of the misalignment. Thus the way that this countermeasure scales the correlation of the correct key (which is our main factor in the “rule of thumb” for the number of needed traces) is not even linear.

## EXAMPLE



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## SECRET SHARING (AKA MASKING)

Masking is based on secret sharing sensitive variables, and computing the entire round function on the shares only.

The security argument for masking consists of two parts:

1. **No leakage below threshold:** Masking schemes typically argue why no reconstruction of the secret is possible below a given threshold. We will look at this part of the security argument shortly.
2. **Exponential cost increase at threshold:** Obviously whenever the leakage of sufficiently many shares is recombined, an attack becomes possible. There is the generic claim that recombination of  $d$  shares increases the effort by an exponential factor of  $d$ .

## ASSUMPTIONS

- ▶ SNR small enough
- ▶ Trace points have roughly equal variance
- ▶ Trace points have roughly equal SNR
- ▶ We use “mean-free product combining” to create joint leakage between trace points.

“Mean-free product combining”: for two trace points  $x$  and  $y$ , compute a new points  $z$  with

$$z = (x - E(x))(y - E(y)).$$

The attack then works on a new trace that is based on points  $z$ , and it proceeds like the attacks that we studied before.

## IMPACT OF SECRET SHARING ON NNT

The variance of the “best”  $z$  is given as

$$\text{Var}(xy) = (\sigma_x^2 + \mu_x^2)(\sigma_y^2 + \mu_y^2) - \mu_x^2\mu_y^2 \geq \sigma_x^2\sigma_y^2.$$

An attack after mean-free product combining of  $d$  points requires

$$n = \frac{1}{\text{SNR}} = \left( \frac{\text{Var}(P_{\text{noise}})}{\text{Var}(P_{\text{exp}})} \right)^d$$

traces. If  $\text{Var}(P_{\text{noise}})$  is the dominating factor (which is our assumption) then this implies an exponential increase by a factor of  $d$ .

## SUMMARY

Countermeasures either diminish the signal (hiding) or, they amplify the noise (secret sharing).

Implementing them correctly is not trivial.

Analysing their effectiveness can be done relatively easily when basing the analysis on correlation.

Countermeasures can drive up the NNT. But if a device has much more signal than noise, then it is virtually impossible to ever get a secure implementation of crypto.